Band-Edge Measurement of Semiconductors
ENPH 352

Prof. David Jones

26th August, 2024

1 Objective

In this experiment you will use a form of optical spectroscopy to measure a few fundamental properties of semiconductors. Specifically, you will be characterizing the absorption edge of semiconductor samples at different temperatures. One application of this technique is the measurement of the process temperature of semiconductors under ultra-high vacuum where other methods fail [1,2]. Absorption band-edge thermometry was studied extensively at UBC, and has now been commercialized.

2 Prelab

Before beginning this prelab, it is suggested that you read through the rest of the lab instructions (below) for context.

In this experiment, the wavelength dependent absorption coefficient α (as defined below in Eqn. 2) will be extracted from optical transmission of light through the sample. However, there are several loss mechanisms inherent in the experimental setup, some of which are also wavelength dependent, that must be taken into account. As an example, the detector used in the spectrometer is silicon-based and Fig. 1 shows a typical spectral response for a silicon detector: clearly not independent of wavelength!

With this in mind, determine an expression for α based on measurement(s) made by the spectrometer. As a first step, sketch out the experimental setup (see Fig. 4). Starting from the light source, trace through the light path to the spectrometer, and note locations where the intensity losses (both wavelength dependent and independent) must be taken into account. For example, the some light will be reflected at the air-sample interface (*hint to quantify this loss, look into Frensel equations from dielectric interfaces.* From this accounting you should be able to find an explicit expression for α which you will use to complete the lab.

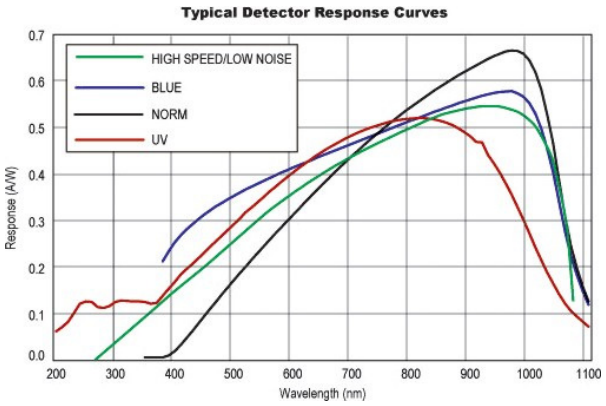


Figure 1: Silicon detector typical response. Note the wavelength dependence as well as different responses depending on the particular optimization (high speed, blue, norm and UV)

3 Theory

3.1 The Semiconductor Bandgap

Crystalline solids may be categorized as conductors or insulators depending on the distribution of electrons [3,4,5]. In a simple picture, the outer electrons in a metal, instead of remaining attached to their home ion, are delocalized over the entire crystal. Electrons in an insulator are more tightly bound. A semiconductor is both a poor conductor and a poor insulator as discussed below.

A more rigorous treatment to quantify these behaviours requires considering electrons not in real space ($\vec{r}$) but in momentum (or wavevector) space, referred to as k-space, as well as quantum and statistical physics. One result of these calculations is a density of states (DOS) which gives the allowable energy states for electrons as a function of energy. A simplified (almost cartoon) representation of DOS for a metal, semiconductor and insulator are shown in Fig. 2.

The energy of the highest occupied electronic state at zero temperature is called the Fermi level, ε_F. If the Fermi energy falls in the middle of a band (i.e. the band is partially-filled) as in left panel of Fig. 2 the crystal is metallic. When this occurs there is always a level slightly higher in energy for the last electron to excite into; hence, in an external field electrons have available states to occupy, they will move and current will flow. The middle panel of Fig.2 display the case of a semiconductor. Within the energy gap, there are no available states for electrons. Thus even under a moderate electric field, there are no allowed energy levels for an electron to occupy from the valence band, hence no current will flow and the material is insulating. For a pure GaAs crystal [4,6], the gap is around 1.4eV; the external field would have to exceed the energy of the bandgap in order to excite electrons to the conduction band and allow current to flow.

A similar situation exists for insulators (the right panel of Fig.2 except that the gap is much larger.

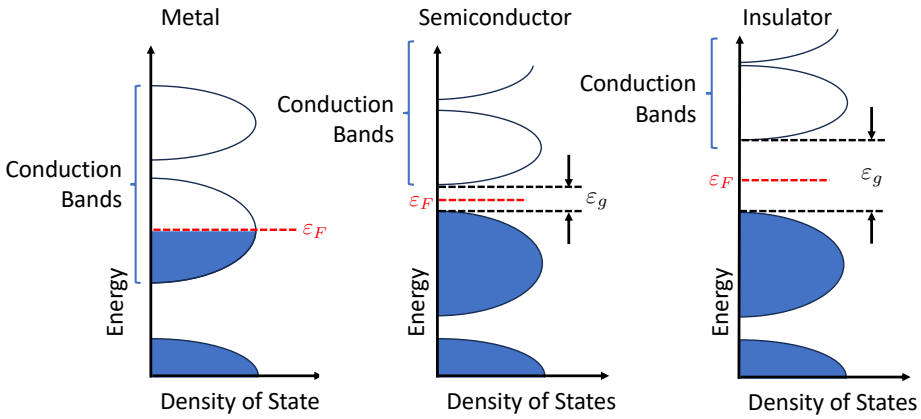


Figure 2: A schematic representation of the density of states of a metal (left panel), semiconductor (middle panel) and insulator (right panel). The metal has a half filled conduction band (up to the Fermi level, ε_F see text) and thus has available states at slightly higher energies for electrons to populate when an electric field is applied. As a result, electrons move and conduction is observed. In contrast, the semiconductor and insulator have an energy gap (ε_g) above the filled valence bands and no electronic states are available to populate under moderate (semiconductor) or high (insulator) electric fields. In these cases electrons can't move and thus they are insulating.

In the above mentioned quantum mechanical analysis of a crystalline solid, an intermediate result before arriving at DOS is the so-called band structure of the material which describes energy levels $\varepsilon(\vec{k})$ where electrons are allowed as a function of their momentum. These bands are combinations of orbitals on all atoms in the crystal, and can only accommodate as many electrons as could be accommodated in their component real-space orbitals. Figure 3 shows part of the band structure of gallium arsenide (GaAs), a semiconductor. As stated previously, for this material, the Fermi level lies somewhere between the "valence" band (downward facing parabola) and the "conduction" band (upward facing parabola), and the highest-energy electron lies at the very top of the valence band. The gap between the valence and conduction bands is the forbidden energy gap, normally called the bandgap and labelled ε_g.

As mentioned above, electrons could be excited from the valence band to the conduction band with a sufficiently strong electric field, but this can also be done with light. If a photon impinges on the crystal with an energy less than ε_g, then it cannot be absorbed and will pass unchanged, just as visible light goes through a window pane. However, if the energy of the photon exceeds ε_g, it can be absorbed by the material, exciting an electron from the valence

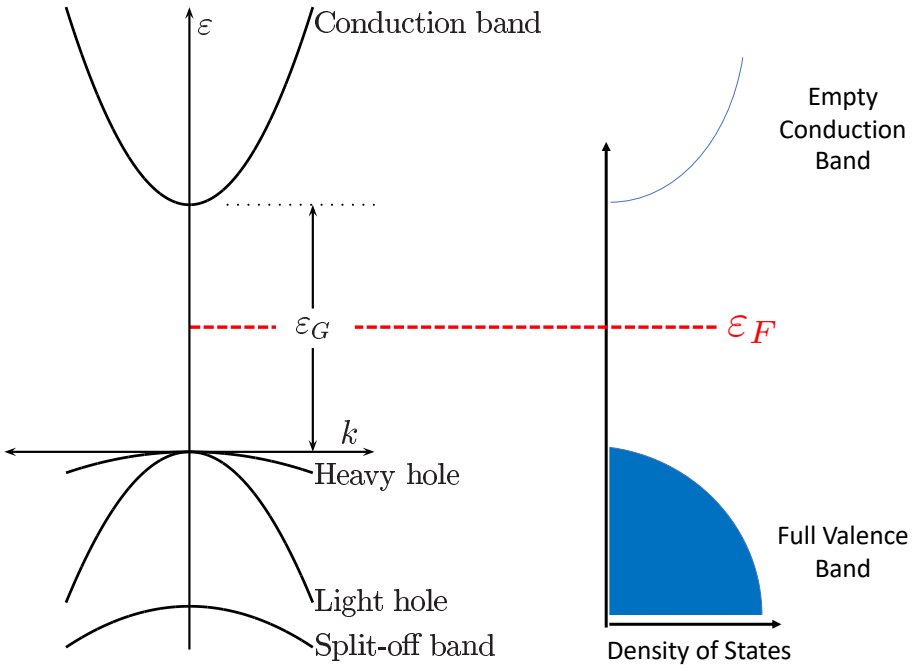


Figure 3: Gallium arsenide (GaAs) near the Γ point (the origin in k-space) — labelled are the conduction band, two valence bands and the split-off band. On the right is a simplified diagram of the electron occupation or density of states . There is no occupation of conduction band and there are no available states above the filled valance bands within a reasonable energy (there is a band gap); rather only when an electron gains enough energy (i.e. on the order of a several kT.) does GaAs become conducting, Accordingly, GaAs is known as a semiconductor.

band into the conduction band, and the material will be opaque to light of that particular color (this is why regular window glass protects us from ultraviolet light). Hence, an experiment examining the absorption of light as function of its wavelength (or energy) can determine the bandgap of a material — the material would be completely transparent below ε_g and completely absorbing above ε_g, leading to a step function absorption spectrum. Remembering the energy of a photon can also be uniquely expressed in terms of its wavelength or frequency, this effect can be summarized as follows:

$$\begin{aligned} h\nu \geq \varepsilon_g; \;\; \alpha \geq \alpha_g \\ h\nu < \varepsilon_g; \;\; \alpha \to 0 \end{aligned} \tag{1}$$

where h is Planck's constant, ν is the frequency of incident light, α is the optical absorption coefficient in cm^{-1} and α_g is the optical absorption coefficient at the band-gap energy. For GaAs, $\alpha_g = 8000\text{cm}^{-1}$ [7].

The preceding discussion is strictly rigorous for a perfect, infinite crystal at absolute zero. In this experiment, however, we will be dealing with real crystals which are finite and have imperfections, and the experiments will be done near room temperature. Both properties contribute to a broadening of the step function, leading to an absorption band-edge known as the Urbach edge [1,2].

The absorption in the Urbach region can be described by:

$$\alpha(h\nu) = \alpha_g e^{\frac{h\nu - \varepsilon_G}{\varepsilon_o}} \tag{2}$$

where ε_o is the characteristic energy (or decay) of the Urbach edge, ε_G is the extrapolated optical bandgap energy, and α_g is again the optical absorption coefficient at the bandgap energy. Note that we denote the Urbach edge optical bandgap with a capital G subscript to distinguish it from the other definition of the bandgap; in practice, we can consider the two quantities to be the same for the purposes of this laboratory, i.e. $\varepsilon_G = \varepsilon_g$. Note this equation is **only** valid for $h\nu < \varepsilon_G$. In quantum theories of the Urbach edge for crystalline semiconductors, both ε_o and ε_G are proportional to the phonon population — phonons are quantized acoustic waves (vibrations of the crystal lattice). Using the Einstein model for phonons, the width of the Urbach edge is

$$\varepsilon_o = S_o k_B \theta_E \left[\frac{1+X}{2} + \frac{1}{e^{\frac{\theta_E}{T}} - 1} \right] \tag{3}$$

where the dimensionless constant X is a measure of the structural disorder, θ_E is the Einstein temperature which corresponds to the average frequency of phonon excitations, S_o is a dimensionless constant related to the electron-phonon coupling, and k_B is the Boltzmann constant. X is expected to be zero for a perfect crystal.

The temperature dependence of the band gap is given by

$$\varepsilon_G(T) = \varepsilon_G(0) - S_g k_B \theta_E \left[\frac{1+X}{2} + \frac{1}{e^{\frac{\theta_E}{T}} - 1} \right] \tag{4}$$

where S_g is a dimensionless coupling constant and $\varepsilon_G(0)$ is taken as the published band gap at liquid-He temperature.

4 Experimental Apparatus

The experimental setup is shown in Fig. 4. A white light source is collimated, passed through a temperature controlled copper block and then directed into a spectrometer (via optical fiber). Different samples (within copper holders) can be screwed on the front of the copper block thereby enabling the spectrometer to measure the wavelength dependent absorption. Aside from the light source and collimating lenses, the apparatus also consists of a diffraction grating spectrometer with a CCD detector (Ocean Optics USB2000) connected to a computer

by USB; a type K thermocouple for measuring the sample's temperature, and a temperature controller that varies current passed through the copper block to maintain a constant temperature.

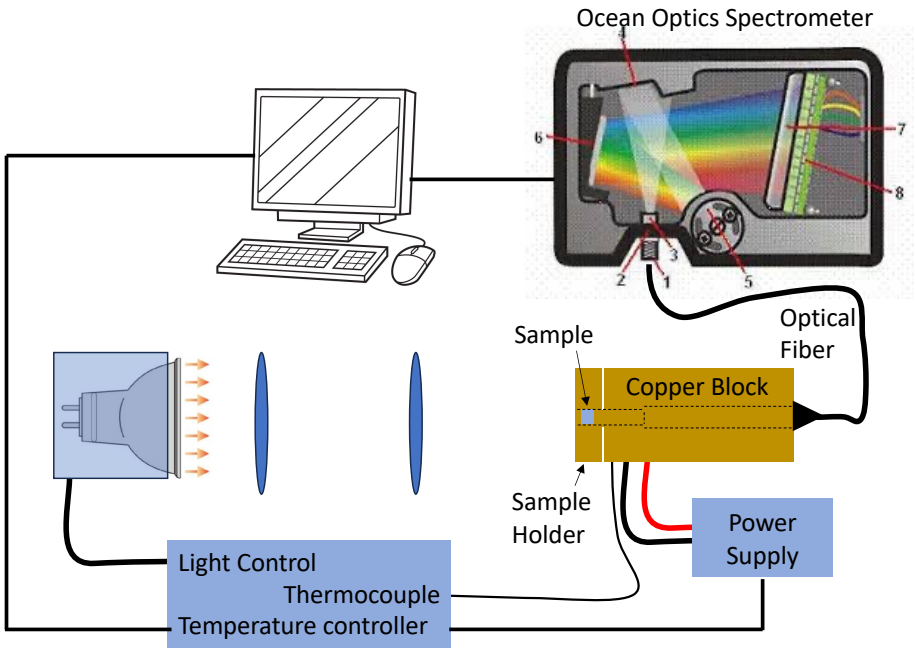


Figure 4: Schematic diagram of the apparatus for your experiment. Both the spectrometer and temperature controller are controlled with LabView programs running on the PC. The temperature controller measures the temperature of the copper block via a thermouple and changes the current passing through the block to stabilize the temperature to a set value.

5 Procedure

5.1 Task #1: Preliminary Measurements

To make sure you collecting data properly, first measure the optical transmission of the provided optical filter. You should make sure you measure the same absolute transmission (both in terms of filter width, center frequency, and maximum transmission) as it specified by the sheet in the lab. To run the spectrometer, use the desktop icon for the Ocean Optics spectrometer (SpectralSuite). Please save your data in `My_Documents\PHYS352\[Yourname]`.

Some possible things to examine and/or understand:

- What spectrum is detected without incoming light (i.e. dark signal)?

- With directly incident light, what is the effect of time normalization? What is the effect of averaging over several curves?
- There are 4 icons in the software labelled `S`, `A`, `T`, and `I` — explain what they stand for and how you can access the `A`, `T` and `I` functions.

5.2 Task #2 — Bandgap of GaAs from Absorption Measurements

- At room temperature, measure the optical transmission through the GaAs sample and then deduce the bandgap and Urbach decay. Note the thickness of GaAs sample can be determined by measuring the thicknesses of scrap GaAs wafers.
- Repeat for different GaAs wafer temperatures. Note that temperature can be controlled using the CNi16D Labview program. **Do not exceed 120 °C**. Does the temperature dependence behave as expected? Hint: refer to references 1 and 2.
- As an extension, you likely have assumed the index of refraction of GaAs is constant. This is actually not the case. To account for its variation with wavelength, you may model the absorption with a Lorenzian lineshape and through Kramers-Kronig relations determine the change in refractive index. See how this refinement impacts your determination of the Urbach decay and overall bandgap.

6 References

1. Shane R. Johnson, *Optical Bandgap Thermometry In Molecular Beam Epitaxy*, Ph.D. thesis, UBC, 1995.
2. M. Beaudoin, A.J.G. DeVries, S.R. Johnson, H. Laman, and T. Tiedje, Appl. Phys. Lett. **70**, 3540 (1997).
3. See for instance: Charles Kittel, *Introduction to Solid State Physics* (Wiley, 2005).
4. K. Seeger, *Semiconductor Physics — An Introduction, 8th Ed.* (Springer-Verlag, 2002), particularly chapters 1 and 2 and sections 11-1 and 11-2.
5. For an advanced treatment (beyond the scope of this course) see: Neil W. Ashcroft and N. David Mermin, *Solid State Physics* (Saunders College, 1976).
6. To make semiconductors useful, impurities are incorporated (doped) into the crystal, in order to make them n-type (conduction by electrons in the conduction band) or p-type (conduction by "holes" or absence of electrons in the valence band). Consult references 1-2 for more information.

7. J.I. Pankove, Phys. Rev. **140**, A2059 (1965)